

Agent-Based Systems: Decoding Their Rise and Business Value

An AI agent can be described as a system that acts on behalf of a user or organisation by taking an active role in achieving a specific goal. Unlike traditional software systems that mainly follow predefined instructions, AI agents can analyse information, make decisions, and interact with their environment to complete tasks. Large language models (LLMs), such as ChatGPT, represented an important development towards this approach, as they allowed users to communicate objectives through natural language and receive assistance in completing tasks. However, despite their initial impact, early LLM-based systems also revealed several limitations, including hallucinations, outdated information, limited context understanding, and difficulties in producing consistently reliable outputs.

The rise of agent-based systems has been driven by recent developments aimed at overcoming these limitations. Techniques such as Retrieval-Augmented Generation (RAG), prompt engineering, fine-tuning approaches, and the integration of external tools have expanded the capabilities of AI agents by allowing them to access additional information and perform more complex tasks. Furthermore, multi-agent systems have increased the potential of this approach by enabling different agents to collaborate and divide responsibilities when solving complex problems (Anthropic, 2024). Agent-based modelling has also been applied to complex systems where interactions between individual agents can help analyse and understand wider patterns and outcomes (Nature Index, n.d.).

As Bonnmann highlights, the main challenge for organisations is no longer whether AI agents will become valuable, but how quickly organisations can create the right conditions for their adoption (Bonnmann, 2022). This reflects the increasing interest in using

artificial intelligence not only for automation, but also as a tool to support decision-making and problem-solving.

For organisations, agent-based systems offer several benefits, including increased efficiency, scalability, and the automation of complex decision-making processes. A practical example can be seen in AI-powered investment platforms such as Wealthfront, which use automated agents to create personalised investment strategies based on an individual's preferences, risk tolerance, and financial objectives. Instead of simply selecting investments with the highest potential return, the system evaluates factors such as market conditions, diversification, and risk management to identify a portfolio that maximises long-term value for the user (Domo, 2026).

However, organisations must also consider the limitations of these systems. Issues such as hallucinations, biased outputs, and unreliable decisions demonstrate that human evaluation and supervision remain essential (Massachusetts Institute of Technology Media Lab, 2025). Therefore, the value of agent-based systems lies not in replacing human expertise, but in supporting human decision-making and enabling organisations to operate more effectively.

References

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